

Summary – Week 3 – Property Dataset

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Feature Selection

In this exercise, I implemented multiple linear regression models using different combinations of the available input features: **Square_Footage**, **Num_Bedrooms**, **Num_Bathrooms**, **Lot_Size**, and **Year_Built**, with **Price** as the target variable. I began with a simple model using only **Square_Footage** because it is generally considered the most influential factor affecting house prices. I then gradually added additional features, including the number of bedrooms, number of bathrooms, lot size, and year built, to evaluate whether each feature contributed to improving the model's predictive performance. This incremental approach allowed me to observe the individual and combined impact of each feature on the prediction accuracy.

Observations from Model Comparisons

The performance of each model was evaluated using **Mean Squared Error (MSE)**, **Root Mean Squared Error (RMSE)**, and the **R-squared (R^2) metric**. The results showed a steady improvement as more relevant features were added. Model 1, which used only **Square_Footage**, achieved an R^2 score of **0.9927**, indicating that house size alone explains a large proportion of the variation in house prices. Adding **Num_Bedrooms** increased the R^2 score to **0.9937**, while including **Num_Bathrooms** further improved it to **0.9939**. The addition of **Lot_Size** produced a slight improvement, and the final model, which included all five features, achieved the highest R^2 score of **0.9941** with the lowest MSE and RMSE values. These results indicate that each additional feature contributed to improving the model, although the improvements became smaller as more variables were added.

Challenges Faced and Solutions

One of the main challenges was determining the most suitable combinations of features for building multiple regression models while ensuring a fair comparison between them. To address this, I used an incremental feature selection strategy, where one feature was added at a time. Another challenge was preparing the data for machine learning using PySpark. This involved assembling multiple feature columns into a single feature vector and applying feature scaling using **StandardScaler** before training the models. Additionally, maintaining the same train-test split for every model ensured that the evaluation results were consistent and directly comparable.

Insights Gained

This exercise enhanced my understanding of multiple linear regression and demonstrated the importance of feature selection in predictive modelling. The results showed that although **Square_Footage** is the strongest predictor of house price, incorporating additional property characteristics improves the overall performance of the model. Comparing multiple models using the R^2 metric illustrated how different combinations of input features affect prediction accuracy. Furthermore, I gained practical experience using PySpark's machine learning pipeline, including feature engineering, model training, evaluation, and comparison. Overall, this exercise strengthened my understanding of regression analysis and highlighted the value of selecting informative features to build accurate predictive models.